

- 3 (a) Define the term *angular velocity*.

Angular velocity is the rate of change of angular displacement.

[1]

- (b) A 10 kg baggage is left on a rotating baggage carousel at an airport as shown in Fig. 3.1.

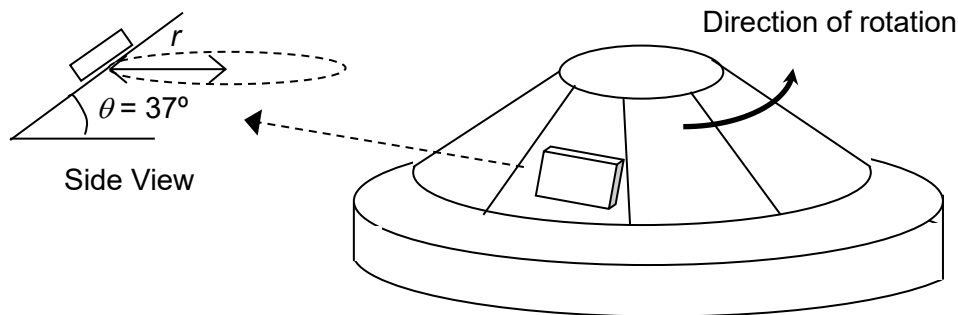


Fig. 3.1

The baggage stays at a fixed position on the slope of the carousel and rotates about in a circle of radius 10 m. The angle θ that the slanted surface makes with the horizontal is 37° . The frictional force acting on the baggage is 60 N. The baggage is moving in uniform circular motion.

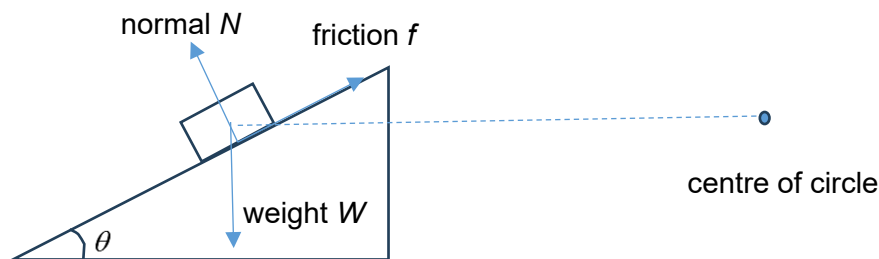
- (i) Explain, using Newton's law(s) of motion, why the baggage will experience a net force towards the centre of the circle.

As the baggage is rotating at constant speed, according to Newton's First Law, a net force is needed to change the direction of motion. [1]

This change in direction of motion results in a rate of change of velocity (acceleration) directed towards the centre of rotation. By Newton's Second Law, the net force acts towards the centre of circle. [1]

[2]

- (ii) Considering the forces acting on the baggage, show that the normal contact force is 78 N.



Consider forces perpendicular to slant surface,

$$N = W \cos \theta \quad [1]$$

$$= (10 \times 9.81) \cos 37^\circ$$

$$= 78.346 \quad [1]$$

$$= 78 \text{ N (to 2 s.f.)}$$

[2]

- (iii) Calculate the time required for the baggage to complete one full rotation.

$$\text{Net force on baggage} = f \cos \theta - N \sin \theta \quad [1]$$

$$mr\omega^2 = f \cos \theta - N \sin \theta$$

$$mr \left(\frac{2\pi}{T} \right)^2 = f \cos \theta - N \sin \theta \quad [1]$$

$$T = 2\pi \sqrt{\frac{mr}{f \cos \theta - N \sin \theta}}$$

$$= 2\pi \sqrt{\frac{(10)(10)}{(60) \cos 37^\circ - (78) \sin 37^\circ}}$$

$$= 64 \text{ s} \quad [1]$$

time = s [3]

[Total: 8]

- 4 (a) Explain what is meant by an *ideal gas*